MSK Embryology — NBME Study Guide

Cartilage & bone formation · Long bone growth · Vertebrae & bone disorders · Skull, sutures & fontanelles · Limb development · Muscle development · Integumentary system

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Origins of the musculoskeletal system

Source	Gives rise to
Paraxial mesoderm → somites (from week 3; ~42–44 pairs)	**Sclerotome → vertebrae, ribs, part of occipital bone; myotome → skeletal muscle of trunk & limbs; dermatome → dermis of back**
Lateral plate mesoderm (somatic/parietal layer)	**Limb bones, cartilage, tendons, ligaments, sternum, dermis of limbs/body wall**
Neural crest	**Most of the face and skull vault (viscerocranium & frontal/parietal parts), melanocytes, Schwann cells**
Notochord	**Nucleus pulposus of intervertebral disc; induces neural tube & sclerotome**
Surface ectoderm	Epidermis, hair, nails, glands, AER of limb bud

2. Cartilage & bone formation

- Mesenchyme condenses → chondroblasts (SOX9) or osteoblasts (**RUNX2/CBFA1**, osterix).
- **Intramembranous ossification: mesenchyme → bone directly (no cartilage) — flat bones of the skull vault, face, mandible, clavicle (also first bone to ossify, week 5–6).**
- **Endochondral ossification: hyaline cartilage model replaced by bone — long bones, vertebrae, pelvis, base of skull.** Steps: cartilage model → chondrocytes hypertrophy & matrix calcifies → periosteal bone collar → nutrient artery invades (periosteal bud) → **primary ossification center in diaphysis** (by week 12) → **secondary centers in epiphyses** (mostly after birth; distal femur at birth) → epiphyseal plates remain until puberty ends.
- Cartilage growth: interstitial (from within) and appositional (from perichondrium).
- **Cleidocranial dysplasia**: RUNX2 mutation (AD) → absent/hypoplastic clavicles (shoulders meet in midline), delayed fontanelle closure, wide sutures, supernumerary teeth — intramembranous bones affected.

3. Long bone growth

- **Length = epiphyseal (growth) plate. Width = appositional growth from periosteum (osteoblasts add, osteoclasts remodel inner surface).**
- **Growth plate zones (epiphysis → diaphysis): reserve/resting → proliferative (columns, stacked coins) → hypertrophic (maturation) → calcified cartilage → ossification.**
- Regulators: GH/IGF-1 (proliferation), thyroid hormone, **estrogen closes the plates** (in both sexes — aromatase/estrogen receptor deficiency → tall stature, open plates), PTHrP/Indian hedgehog loop, FGFR3 (inhibits proliferation).
- **Achondroplasia**: activating FGFR3 mutation (AD; most cases new, ↑ paternal age) → ↓ chondrocyte proliferation in the growth plate → short limbs, normal trunk, large head, frontal bossing, normal intelligence; intramembranous bone normal.
- Rickets (vitamin D deficiency, children): defective mineralization at the hypertrophic zone → widened plates, bowed legs, rachitic rosary, craniotabes. Salter–Harris fractures involve the plate.
- **Gigantism** (↑ GH before plate closure) vs **acromegaly** (after closure — only appositional growth: hands, feet, jaw).

4. Vertebrae & bone disorders

- **Resegmentation**: each sclerotome splits; the caudal half of one fuses with the cranial half of the next → vertebral body is **intersegmental**; spinal nerves exit between vertebrae; myotomes bridge discs (so muscles move the spine).

- **Intervertebral disc: nucleus pulposus = notochord remnant; annulus fibrosus = sclerotome (fibrocartilage).** Herniation = nucleus through annulus (posterolateral).
- Neural arch closure defects: **spina bifida occulta** (failure of vertebral arch fusion, no herniation; tuft of hair/dimple at L5–S1; normal AFP), **meningocele** (meninges herniate), **myelomeningocele** (meninges + cord; ↑ AFP; Arnold–Chiari II). Folate prevents NTDs.
- **Klippel–Feil syndrome**: fusion of cervical vertebrae → short neck, low hairline, limited neck motion.
- **Hemivertebra** (failure of half a vertebra to form) → congenital scoliosis. Scoliosis (lateral), kyphosis (thoracic hump), lordosis (lumbar).
- Cervical rib (from C7) → thoracic outlet syndrome. Lumbar rib, sacralization of L5 / lumbarization of S1.
- Primary curvatures (thoracic, sacral — fetal) vs secondary (cervical when holding head up, lumbar when walking).
- Other congenital bone disorders: **osteogenesis imperfecta** (type I collagen, blue sclerae, fractures), **osteopetrosis** (osteoclast failure, dense brittle bones), **Marfan** (fibrillin-1), **arthrogryposis** (multiple joint contractures from ↓ fetal movement/oligohydramnios).

5. Skull, sutures & fontanelles

- **Neurocranium** (encloses brain): **membranous part** = flat bones of the vault (intramembranous; neural crest & paraxial mesoderm); **cartilaginous part (chondrocranium)** = base of skull (endochondral). **Viscerocranium** = face (mainly neural crest from pharyngeal arches 1–2).
- Sutures (dense CT) allow molding at birth and brain growth. **Anterior fontanelle (diamond; frontal–parietal junction) closes ~18–24 months; posterior fontanelle (triangular) closes ~2–3 months.** Bulging fontanelle → ↑ ICP; sunken → dehydration.
- **Craniosynostosis** = premature suture closure → skull grows parallel to the fused suture:
  - **Sagittal (most common, males) → scaphocephaly / dolichocephaly (long, narrow head).**
  - **Coronal** (bilateral) → brachycephaly (short, wide; Apert/Crouzon — FGFR2); unilateral → anterior plagiocephaly.
  - **Lambdoid** (unilateral) → posterior plagiocephaly (vs positional plagiocephaly from sleeping position — far more common).
  - **Metopic** → trigonocephaly (triangular forehead).
  - Multiple sutures → oxycephaly/turricephaly (tower skull), cloverleaf skull (FGFR).
- Microcephaly (small brain → small skull; Zika, alcohol) vs anencephaly (failure of anterior neuropore — no vault).

6. Limb development

- **Limb buds appear end of week 4 (upper slightly before lower); key period weeks 4–8** — most limb defects (e.g., **thalidomide → phocomelia/amelia**, days 24–36).
- Bud = lateral plate mesoderm core (bones, CT) covered by ectoderm; **muscle from somite myotomes** migrates in (dorsal/ventral muscle masses → extensors/flexors).

Signaling center	Molecule	Axis / role
Apical ectodermal ridge (AER)	**FGFs (FGF8)**	**Proximal–distal outgrowth; removal → truncated limb**
Zone of polarizing activity (ZPA), posterior	**Sonic hedgehog (SHH)**	**Anterior–posterior (thumb ↔ little finger); ectopic anterior ZPA → mirror duplication / polydactyly**
Dorsal ectoderm	WNT7a	Dorsal–ventral (nails dorsal)
HOX genes	—	Segment identity; mutations → limb malformations (synpolydactyly)

- **Digits separate by apoptosis** in interdigital regions (weeks 6–8; BMPs) → failure = **syndactyly**. Extra digits = polydactyly.
- **Limb rotation (week 7)**: upper limb rotates **90° laterally** (extensors posterior, thumb lateral); lower limb rotates **90° medially** (extensors anterior, big toe medial) → dermatomes spiral in the leg.
- Defects: amelia (absent limb), meromelia (partial), phocomelia (hands/feet attached close to trunk), clubfoot (talipes equinovarus — most common; inverted, plantarflexed), congenital hip dysplasia, amniotic band syndrome (constriction rings/amputations).

- Joints: interzones of mesenchyme → synovial joints (cavity by apoptosis); fibrous and cartilaginous joints form where mesenchyme differentiates.

7. Muscle development

- **Skeletal muscle: from paraxial mesoderm (somite myotomes)**; head muscles from somitomeres/pharyngeal arch mesoderm. Myoblasts (MyoD, Myf5, myogenin) **fuse** → multinucleated myotubes → fibers; satellite cells remain as stem cells.
- **Epimere** (dorsal) → deep back extensors (dorsal rami); **hypomere** (ventral) → body wall & limb muscles (ventral rami).
- **Smooth muscle: splanchnic (visceral) lateral plate mesoderm** (gut, vessels); iris sphincter/dilator from neuroectoderm. **Cardiac**: splanchnic mesoderm around the heart tube.
- **Poland sequence**: absent pectoralis major (sternocostal head) ± ipsilateral syndactyly/breast hypoplasia. **Prune belly** syndrome: absent abdominal wall muscles. Congenital torticollis (SCM fibrosis).
- **Duchenne/Becker** (dystrophin) — X-linked; Gowers sign; covered in histology/cell bio.

8. Integumentary system

- **Epidermis = surface ectoderm; dermis = mesoderm (dermatome on back, lateral plate elsewhere); melanocytes = neural crest** (migrate by weeks 8–12). Langerhans cells = bone marrow; Merkel cells = neural crest.
- Epidermis starts as a single layer + **periderm** (shed into amniotic fluid → **vernix caseosa** with sebum); stratified by ~week 21.
- **Hair follicles**: epidermal downgrowth into dermis; **lanugo** (fine fetal hair, shed before/after birth) replaced by vellus/terminal hair. Hair papilla = mesoderm.
- Glands (all from epidermis): sebaceous (from follicle wall), sweat glands, **mammary glands** (modified sweat glands along the milk line → supernumerary nipples = polythelia).
- Nails start ~week 10 at fingertips (reach tips ~32 weeks; toes later — useful for gestational age).
- Teeth: enamel = ectoderm (ameloblasts); dentin/pulp = neural crest (odontoblasts).
- Disorders: **albinism** (normal melanocyte number, ↓ tyrosinase/melanin), vitiligo (↓ melanocytes, autoimmune), **ichthyosis** (abnormal keratinization; harlequin), **ectodermal dysplasia** (absent sweat glands, sparse hair, abnormal teeth), epidermolysis bullosa (anchoring proteins: keratin, collagen VII), hemangiomas.

9. Rapid-fire NBME pearls

- **Intramembranous: skull vault, face, clavicle. Endochondral: long bones, vertebrae, skull base.**
- **Achondroplasia = FGFR3 (proliferative zone); cleidocranial dysplasia = RUNX2.**
- **Nucleus pulposus from notochord; vertebrae from sclerotome by resegmentation.**
- **Sagittal synostosis → scaphocephaly (long narrow head); anterior fontanelle closes by ~18–24 months.**
- **AER = FGF (length); ZPA = SHH (anterior–posterior, polydactyly); digits separate by apoptosis (syndactyly).**
- **Upper limb rotates laterally, lower limb medially; limb buds week 4; thalidomide → phocomelia.**
- **Skeletal muscle = myotome (paraxial mesoderm); smooth = splanchnic mesoderm; melanocytes = neural crest.**